

Cryptology Assignment - 5

Rail Fence:

1. Encrypt the message "Enjoy your journey" using Rail Fence with depth 2 and depth 3:

Ans: Depth 2: Plaintext: Enjoyyourjourney
Ciphertext?

```

E   J   Y   O   R   O   R   E
  N   O   Y   U   J   U   N   Y
  
```

Ciphertext = EJYORORENOYUJUNY

Depth 3:

```

E       Y       R       R
  N   O   Y   U   J   U   N   Y
    J       O       O       E
  
```

Ciphertext: EYRRNOYUJUNYJOOE

2. Encrypt the message as per your choice using Rail Fence with depth 2 and 3.

Ans: Plaintext: The sky is pink.
⇒ The sky is pink.

Depth 2:

```

T   E   K   I   P   N
  H   S   Y   S   I   K
  
```

Ciphertext: TEKIPNHSYSIK.

Depth 3:

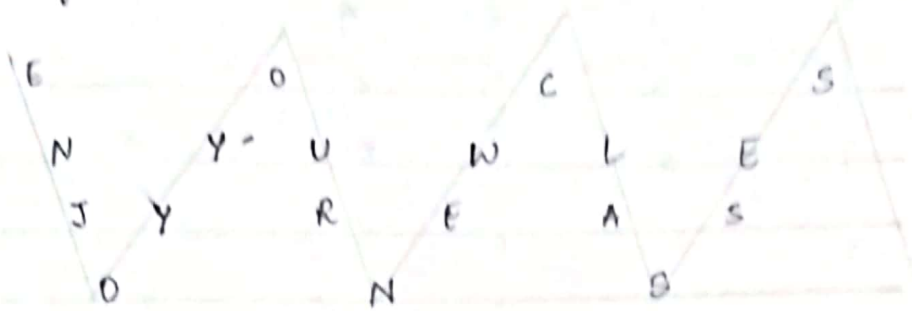
```

T       K       P
  H   S   Y   S   I   K
    E       I       N
  
```

Ciphertext: TKPHSYSIKEIN

3. Decrypt the cipher "EOCSNYUWLEJYREASNS", encrypted using Rail fence with depth 4:

Ans:

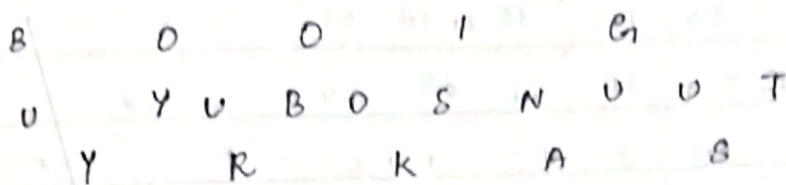


Plaintext: ENJOY YOUR NEW CLASSES.

"Enjoy your new classes"

4. Decrypt the cipher "BODUGUYUBOSNUUJYRKAS", encrypted using Rail Fence with depth 3.

Ans:

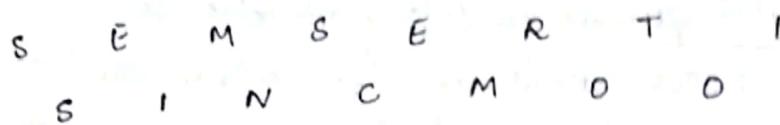


Plaintext: BUY YOUR BOOKS IN AUGUST.

"Buy your books in August"

5. Decrypt the cipher "SEMERTISINCMOO" encrypted using Rail Fence with depth 2.

Ans:



Plaintext: SEIMNSCEMROTOI.

Permutation:

1. Encrypt the following message "We are discovered" using the key "53124".

Ans: Message: We are discovered

→ Wearediscovered

Key: 53124.

①	②	③	④	⑤
5	3	1	2	4
w	e	a	n	e
d	i	s	c	o
v	e	n	e	d

Order: ③, ④, ②, ⑤, ①

a s n e e e e o d w d v → ciphertext.

BES

Initial Permutation:

58	50	42	34	26	18	10	02
60	52	44	36	28	20	12	04
62	54	46	38	30	22	14	06
64	56	48	40	32	24	16	08
57	49	41	33	25	17	09	01
59	51	43	35	27	19	11	03
61	53	45	37	29	21	13	05
63	55	47	39	31	23	15	07

- Using the Initial permutation table, determine the output of the Initial permutation box when the input is given in hexadecimal as:

0x0002 0000 0000 0001

Ans: Converting them to the binary

0000 0000 0000 0001 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0001

Only two bits are '1': Input bits 15 and 64.

Every other are 0.

64 appears in column 1, Row 4 $\rightarrow$ that is 35th entry in the table $\rightarrow$ output bit 25 = input bit 64 = 1

15 appears in Row 8, column 7 $\rightarrow$ that is the 63rd entry in the table $\rightarrow$ output bit 63 = input bit 15 = 1

All other output bits = 0

0000 0000 0000 0000 0000 0000 1000 0000 0000 0000 0000
0000 0000 0000 0000 0000 0000 .

Output:

010000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000

2. Given the 4-bit binary plaintext (M) 1101 and the Key (K) 1011, calculate the resulting Ciphertext (C) using a bitwise Exclusive-OR (XOR) operation. Show your bit-by-bit work. Plaintext (M): 1101 (binary) · Key : 1011 (binary)

Ans: XOR of M = 1101 and K = 1011

$$M = 1101$$

$$K = 1011$$

$$\begin{array}{r} 1101 \\ 1011 \\ \hline 0110 \end{array}$$

$$C = \underline{0110}$$

3. The input to S-box 1 is 000000. What is the output?

14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

Product cipher:

Ans: Input = 000000

Check the 0th position - 14.

$$14 = \underline{1110}$$

$$\begin{array}{r} 2 \overline{) 14} - 0 \\ 2 \overline{) 7} - 1 \\ 2 \overline{) 3} - 1 \\ 1 \end{array}$$